

Amazon 데이터셋 기반 추천 시스템 설계 프로젝트 - 허재민

결측치 제거 및 카테고리 split

```
-- clean 데이터

CREATE OR REPLACE TABLE `project-23e56097-5f6c-4899-bd8.project2.amazon_clean` AS
SELECT
  * EXCEPT(rating),
  CAST(rating AS FLOAT64) AS rating
FROM `project-23e56097-5f6c-4899-bd8.project2.amazon`
WHERE
  rating_count IS NOT NULL
  AND rating IS NOT NULL
  AND rating != '|';

-- clean + split 데이터

CREATE OR REPLACE TABLE `project-23e56097-5f6c-4899-bd8.project2.amazon_user_split` AS
SELECT
  product_id,
  product_name,
  discounted_price,
  actual_price,
  discount_percentage,
  CAST(rating AS FLOAT64) AS rating,
  rating_count,
  about_product,
  category,
  SPLIT(category, '|')[SAFE_OFFSET(0)] AS category_1,
  SPLIT(category, '|')[SAFE_OFFSET(1)] AS category_2,
  SPLIT(category, '|')[SAFE_OFFSET(2)] AS category_3,
  SPLIT(category, '|')[SAFE_OFFSET(3)] AS category_4,
  SPLIT(category, '|')[SAFE_OFFSET(4)] AS category_5,
  user_id_arr AS user_id,
  SPLIT(user_name, ',')[SAFE_OFFSET(pos)] AS user_name,
  review_id,
  review_title,
  review_content
FROM `project-23e56097-5f6c-4899-bd8.project2.amazon`,
UNNEST(SPLIT(user_id, ',')) AS user_id_arr WITH OFFSET pos
WHERE
  rating_count IS NOT NULL
  AND rating IS NOT NULL
  AND rating != '|';
```

레시피 1: "최대 할인가 제품 추천 시스템"

컨셉: 할인율이 높다고 가격을 많이 깎아주는것은 아닙니다, 큰 금액을 깎아주는 제품을 추천 함으로 고객들은 많은 돈을 절약할 수 있다고 생각합니다. (총 할인가를 제공)

핵심 질문: 할인율이 높다고 가격적인 측면에서 매력적인 것은 아니지 않나?

주요 재료: `rating` (평점), `actual_price` (정가), `discounted_price` (할인 된 가격)

```
SELECT
  product_id,
  product_name, actual_price - discounted_price AS discount_amount,
  rating
FROM `project-23e56097-5f6c-4899-bd8.project2.amazon_clean`
WHERE rating >= 4.5
```

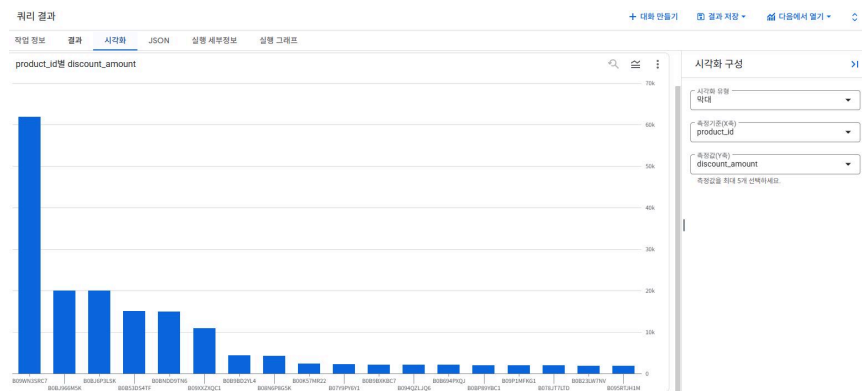
```
ORDER BY discount_amount DESC, rating DESC, rating_count DESC
LIMIT 20 ;
```

출력 결과:

쿼리 결과 + 대화 만들기 결과 저장

작업 정보	결과	시각화	JSON	실행 세부정보	실행 그래프
명	product_id	product_name	discount_amount	rating	
1	B09WN3SRCT	Sony Bravia 164 cm (65 inches) 4K Ultra HD Smart LED Google TV KD-65X74K (Black)	61910.0	4.7	
2	B0BJ966MSK	Aquapure Copper + Mineral RO+UV+UF 10 to 12 Liter RO + UV + TDS ADJUSTER Water Purifier with Copper Charge Technology black & copper	20000.0	4.6	
3	B0BJ9P3LSK	Aqua d pure Active Copper 12.4 L RO+UV Water Filter Purifier for Home, Kitchen Fully Automatic UF+TDS Controller	20000.0	4.5	
4	B0BS5DS4TF	Instant Pot Air Fryer, Vortex 2QT, Touch Control Panel, 360° EvenCrisp™ Technology, Uses 95 % less Oil, 4-in-1 Appliance: Air Fry, Roast, Bake, Broil, Dehydrate	15054.0	4.8	
5	B0BNDD9TN6	WANBO X1 Pro (Upgraded) Native 1080P Full HD Android 9 Projector for Home LED Cinema 350ANSI	14910.0	4.5	

페이지당 결과 수: 50 1 ~ 20 (전체 20행)



레시피 2: "숨겨진 가성비 제품 추천"

컨셉: 아직 고객들에게 발굴되지 않았지만 가성비가 좋은 아이템을 추천

핵심 질문: 나만의 작은 가성비 제품 어디 없나?

주요 재료: `rating` (평점), `rating_count` (평점 갯수), `discounted_percentage` (할인율)

```
WITH b AS (
  SELECT
    APPROX_QUANTILES(rating_count, 4)[OFFSET(1)] AS q1
  FROM `project-23e56097-5f6c-4899-bd8.project2.amazon_clean`
)

SELECT
  product_id,
  product_name,
  rating,
  rating_count,
  discount_percentage
FROM `project-23e56097-5f6c-4899-bd8.project2.amazon_clean`, b
WHERE
  rating_count BETWEEN 500 AND q1
  AND discount_percentage >= 0.5
  AND rating >= 4
ORDER BY discount_percentage DESC, rating DESC;
```

출력 결과:

쿼리 결과

+ 대화 만들기

결과 저장

다음에서 열기

작업 정보

결과

시각화

JSON

설명 세부정보

설명 그래프

id	product_id	product_name	discounted_price	discount_percent	rating	rating_count
1	B0712DN3TQ	Ro"™ High-Speed 3D Full HD 1080p Support (10 Meters) HDMI Male to HDMI Male Cable TV Lead 1.4V for All Hdmi Devices- Black (10M-	598.0	0.88	4.2	910
2	B09T16BFDX	Colbolt Silicone Protective Case Cover for LG an M621GA Magic Remote Shockproof for LG Smart TV Remote 2021 Protective Skin Waterproof	399.0	0.8	4.5	505
3	B08PFSZ7FH	STRIFF Laptop Stand Adjustable Laptop Computer Stand Multi-Angle Stand Phone Stand Portable Foldable Laptop Riser	299.0	0.8	4.2	903
4	B09RZ51NQ7	Source 65W OnePlus Dash Warp Charge Cable, 6.5A Type-C to USB C PD Data Sync Fast Charging Cable Compatible with One Plus 8T/	199.0	0.8	4.0	575
5	B09RZ51NQ7	Source 65W OnePlus Dash Warp Charge Cable, 6.5A Type-C to USB C PD Data Sync	199.0	0.8	4.0	575

페이지당 결과 수: 50

1 - 49 (전체 49명)

페이지당 결과 수: 50 1 ~ 49 (전체 49명) < >

레시피 3: "파워 유저가 선택한 제품 추천"

컨셉: 지금 시장에 큰손들이 주목하는 트렌드 상품을 추천하자, 리뷰 많이 남긴 유저 = 파워 유저 (트렌드 픽)

핵심 질문: 가장 활발히 활동하는 유저들이 찍한 상품은 뭐가있을까? 그리고 그 트렌드(카테고리)는 어떤걸까?

주요 재료: **user_id** (유저 고유 아이디), **user_name** (유저 닉네임), **rating** (평점), **review_cnt** (리뷰갯수)

```
WITH top_reviewers AS (
  SELECT user_id,
         ANY_VALUE(user_name) AS user_name,
         COUNT(*) AS review_cnt
  FROM `project-23e56097-5f6c-4899-bd8.project2.amazon_user_split`
  GROUP BY user_id
  ORDER BY review_cnt DESC
  LIMIT 100
)

SELECT
  a.product_id,
  ANY_VALUE(a.product_name) AS product_name,
  ANY_VALUE(a.category_1) AS category_1,
  ANY_VALUE(a.discounted_price) AS discounted_price,
  AVG(a.rating) AS rating_avg_by_top_reviewers,
  COUNT(*) AS review_cnt_by_top_reviewers
FROM `project-23e56097-5f6c-4899-bd8.project2.amazon_user_split` a
JOIN top_reviewers t
ON a.user_id = t.user_id
GROUP BY a.product_id
HAVING COUNT(DISTINCT a.user_id) >= 5
      AND rating_avg_by_top_reviewers >= 4
ORDER BY rating_avg_by_top_reviewers DESC, review_cnt_by_top_reviewers DESC
```

출력 결과:

쿼리 결과

+ 대화 만들기

결과 저장

다음에서 열기

작업 정보

결과

시각화

JSON

설명 세부정보

설명 그래프

	product_id	product_name	category_1	discounted_price	rating_avg_by_top	review_cnt_by_top
1	B08RVYFKP	SanDisk Ultra microSDXC™ UHS-I Card, 64GB, 140MB/s R, 10 Y Warranty, for Smartphones	Electronics	569.0	4.4	16
2	B08L9HAUJW	SanDisk Ultra microSD UHS-I Ca.	Electronics	369.0	4.4	8
3	B08L9HAUJC	SanDisk Ultra microSD UHS-I Ca.	Electronics	649.0	4.4	8
4	B08DYW39N3	SanDisk Ultra microSDXC™ UHS-I Card, 256GB, 150MB/s R, 10 Y Warranty, for Smartphones	Electronics	1999.0	4.4	8
5	B08DYVCSTD	SanDisk Ultra microSDXC™ UHS-I Card, 128GB, 140MB/s R, 10 Y Warranty, for Smartphones	Electronics	999.0	4.4	8
6	B08HDJ66NZ	boM Deuce USB 300 2 in 1 Type-C & Micro USB Stress Resistant, Tangle-Free, Sturdy Cable with 3A Fast Charging & 480Mbps Data Transmission, Woven Nylon Braided USB to Lightning Fast Charging and Data Sync Cable Compatible for iPhone 13, 12, 11, X, 8, 7, 6, 5, SE, iPad Pro, iPad Air, iPad Mini	Computers&Accessories	329.0	4.2	24
7	B07J9FH4J1	Woven Nylon Braided USB to Lightning Fast Charging and Data Sync Cable Compatible for iPhone 13, 12, 11, X, 8, 7, 6, 5, SE, iPad Pro, iPad Air, iPad Mini	Computers&Accessories	399.0	4.2	24

페이지당 결과 수: 50

1 - 48 (전체 48명)

페이지당 결과 수: 50 1 ~ 49 (전체 49명) < >

레시피 4: "각광받는 1위 ~ 5위 제품 추천"

컨셉: 전체 시장에서 각광 받는 제품 5순위 추천

핵심 질문: 현재 평점높고 리뷰많은 5개 제품이 뭐가 있을까

주요 재료: `rating_count` (평점 갯수), `rating` (평점)

```
WITH base AS (  
  SELECT *  
  FROM `project-23e56097-5f6c-4899-bd8.project2.amazon_clean`  
  ORDER BY rating_count DESC  
  LIMIT 5  
)  
SELECT  
  product_id,  
  product_name,  
  rating,  
  rating_count,  
  discount_percentage,  
  discounted_price  
FROM base  
ORDER BY rating DESC, rating_count DESC
```

출력 결과:

쿼리 결과							
작업 정보 결과 시각화 JSON 실행 세부정보 실행 그래프							
행	product_id	product_name	rating	rating_count	discount_percent	discounted_price	
1	B07K5MBL2H	AmazonBasics Flexible Premium HDMI Cable (Black, 4K@60Hz, 18Gbps), 3-Foot	4.4	426973	0.69	219.0	
2	B014IBSX4Y	Amazon Basics High-Speed HD...	4.4	426973	0.78	309.0	
3	B014IBSSD0	Amazon Basics High-Speed HDMI Cable, 6 Feet - Supports Ethernet, 3D, 4K video, Black	4.4	426973	0.35	309.0	
4	B07K5MBL2H	AmazonBasics Flexible Premium HDMI Cable (Black, 4K@60Hz, 18Gbps), 3-Foot	4.4	426972	0.69	219.0	
5	B07GPKXNNG	boAt Bassheads 100 in Ear Wired Earphones with Mic (Tasty Pink)	4.1	363713	0.65	349.0	

레시피 5: "카테고리 별 Top 1 추천"

컨셉: 각 카테고리 별 최고의 제품 추천

핵심 질문: 대분류 카테고리 중에서 가장 인기상품은 뭘까?

주요 재료: `category_1` (대분류 카테고리), `rating` (평점), `rating_count` (평점갯수),

```
WITH base AS (  
  SELECT  
    category_1,  
    product_id,  
    ANY_VALUE(product_name) AS product_name,  
    AVG(rating) AS rating,  
    ANY_VALUE(rating_count) AS rating_count  
  FROM `project-23e56097-5f6c-4899-bd8.project2.amazon_user_split`  
  WHERE category_1 IS NOT NULL  
  GROUP BY category_1, product_id  
)  
ranked_base AS (  
  SELECT  
    category_1,  
    product_id,  
    product_name,  
    rating,  
    rating_count,  
    ROW_NUMBER() OVER(  
      PARTITION BY category_1
```

```

ORDER BY rating DESC, rating_count DESC
) AS rn
FROM base
)
SELECT
category_1,
product_id,
product_name,
rating,
rating_count
FROM ranked_base
WHERE rn = 1
ORDER BY category_1

```

출력 결과:

쿼리 결과 + 대화 만들기 [결과 저장](#) [다른에서 열기](#)

작업 정보	결과	사라화	JSON	실행 세부정보	실행 그래프
명	category_1	product_id	product_name	rating	rating_count
1	Car&Motorbike	B0912WJ87V	Breffer AX30 (MAX) Portable Air Purifier for Car, Home & Office Smart Ionizer Function H13 Grade True HEPA Filter Internationally Tested	3.8	1118
2	Computers&Accessories	B09ZHCJDP1	Amazon Basics Wireless Mouse 2.4 GHz Connection, 1600 DPI Type-C Adapter Up to 12 Months of Battery Life Ambidextrous Design	5.0	23
3	Electronics	B08Z3LW7NV	Spigen EZ Fit Tempered Glass Screen Protector for iPhone 14 Pro Max - 2 Pack (Sensor Protection)	4.7	7779
4	Health&PersonalCare	B07BK5DR2	Dr. Trust Electronic Kitchen Digital Scale Weighing Machine (Blue)	4.0	3663
5	Home&Kitchen	B08B4FS7BQ	Swiffer Instant Electric Water Heater Faucet Tap Home Kitchen Instantaneous Water Heater Tank less for Tap, LED Electric Head Water	4.8	53803

페이지당 결과 수: 50 • 1-9 (현재 9명) | < >